

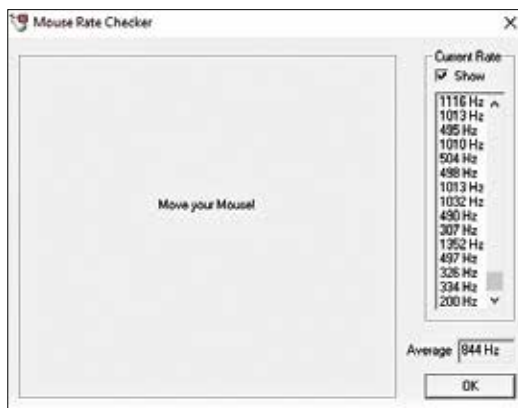
delay or mouse latency of 1ms in the reporting between your PC and mouse.

Why overclock

Even though the latency is as small as 1ms, the total input lag could be much higher. The total input lag is a combination of a whole bunch of factors including the mouse latency, frame rendering time, display latency, etc. If you're using a standard display with a 60Hz refresh rate and a mouse with 125Hz, then you're going to have an input lag of 16.67ms (1/60 sec). With a 120Hz display and the input lag comes down to 8.3ms. If you move to a 144Hz display, the input lag will be 6.9ms and finally on a 240Hz display, the input lag further drops to 4.1ms. So, as you can see as the refresh rate of the display increases, the input lag goes down and eventually the polling rate of the mouse becomes increasingly important. If you use a 125Hz mouse on a 120Hz display, there are high chances that the frame might get updated without any new input from your mouse. This poses a problem when you're playing FPS games and you can't risk losing any sort of mouse input. In this case, you either have the option of buying a new mouse with a higher polling rate or by overclocking the mouse you have. There could be times where you might need to lower the polling rate of your mouse when it starts eating up a lot of CPU resources. It only happens if your CPU isn't powerful enough to simultaneously handle the high reporting rate of the mouse along with other applications or games. However, if you have new generation CPUs, they should be capable of handling a reporting rate of even 1,000Hz while gaming.

Tweaking the polling rate

Before we begin, how do you determine your mouse's polling rate? There are several options out there and to keep things easy, we would recommend Zowie's online tool (<https://dgit.in/ZowieMouseRate>) or Mouse Rate Checker (<https://dgit.in/MRChck>) if you want to use a software. All you need to do is click on the web page and move your mouse for a few seconds to determine your polling rate. This



Verify your mouse's polling rate each time

guide is going to be extremely beneficial to you if you're stuck with a 125Hz mouse since ramping it up to 1,000Hz will make a lot of difference. If you already own a mouse supporting 500Hz or 1,000Hz, you could try to push it beyond to even up to 8,000Hz (highly unlikely to happen). Do remember, not all mice will be capable of crossing the 1,000Hz threshold.



Always connect your mouse to the USB Root Hub (USB 3.0) controller

In order to increase or decrease the polling rate, you'll need to download a USB mouse overclocking tool called the USB Mouse Rate Adjuster from here: <https://dgit.in/hidusbf>. Make sure you've connected the mouse to a USB 3.0 or above port and which has a parent called "USB Root Hub". Extract the *hidusbf* folder and install the signed certificate by running *SweetLow.CER*. Head over to the "DRIVERS" folder and run the setup executable file. All your USB devices will be listed here and your mouse can be filtered out by checking on the 'Mice Only' box. You have to ensure the 'Parent Name' against your device is listed as 'USB Root Hub'. On the bottom, the first dropdown list will let you select the polling rate, right from 31Hz to 1,000Hz. To check whether everything is working, try selecting any one of them. Then hit 'Install service' and then 'Restart'. Hitting the restart button will not work for a few mice, so after installing the service, you will have to reconnect the mouse to make

the changes. Head over and check the polling rate to verify whether it worked or not. For most mice out there, this should work without any sweat. The tricky part comes when you want to further push the polling rate beyond 1,000Hz. For that, you will need to copy the driver from the 2kHz-4kHz folder to your Windows driver directory. After moving, install the service from the

software and reboot your system. Run the software again and select 31 or 62 and check the polling rate. Choosing 31 should ideally give you a polling rate of 2,000Hz while choosing 62 should give you 4,000Hz. If you wish to try your hands at bumping it up more, then follow the same procedure with the driver in the 4kHz-8kHz folder. A more detailed and step by step instruc-

tions are given inside the folder in the *README.2kHz-8kHz.ENG.TXT* file. Even if you can't go beyond 1,000Hz, you will still be able to breathe new life into older mice with low polling rates using this method.

OVERCLOCKING YOUR KEYBOARD

When it comes to keyboards, there isn't any benefit at all. The only kind of input going from your keyboard to your PC is the keystroke data which is relatively slow compared to the mouse data. Realistically speaking, you won't be able to type fast enough to get even close to the reporting rate of the keyboard. Hence, a polling or reporting rate of 125Hz or 1,000Hz won't make any noticeable difference in your typing or gaming sessions. Since there's nothing substantial to achieve from this exercise or witness the upgrade, we will skip overclocking a keyboard. However, if you really want to do it, you can follow the same procedure as that of the mouse.